

We Are Still Here: The Thámien Ohlone Augmented Reality Tour

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ABSTRACT

The Thámien Ohlone Augmented Reality (AR) Tour at Santa Clara University uses an innovative approach to highlight the overlooked story of the Ohlone Native American People on a campus dominated by Catholic monuments. This location-based AR tour integrates 3D models, narration by Native storytellers, and traditional songs at three strategic stops, fostering a deep connection between the land and its indigenous stories. The project follows the principles of "design justice," emphasizing the active participation of the Muwekma Ohlone Tribe and adherence to the concept of rhetorical sovereignty. The "We Are Still Here" video shows novel co-design methods, such as "landmark-based affinity diagramming," that align stories with specific locations. Overall, it showcases how location-based AR can be a powerful medium for social justice narratives.

CCS CONCEPTS

• **Human-centered computing** → *Human computer interaction (HCI)*; **Mixed / augmented reality**; • **Applied computing** → *Arts and humanities*.

KEYWORDS

augmented reality, spatial justice, design justice, social justice

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The campus of Santa Clara University features dozens of Catholic monuments and is home to the Mission Santa Clara, an active Roman Catholic Church. Yet the presence of the Ohlone Native American People, who have lived on the land for thousands of years, is largely unrecognized at present. In 2020, a working group of Ohlone leaders, scholars, and university administrators offered a list of recommendations to more fully acknowledge the Native history of our campus [1], but the process of creating new physical interpretive markers and displays has yet to begin in earnest. To address this erasure, the Thámien Ohlone Augmented Reality (AR) Tour is crafted to reveal hidden histories, spur visitors towards critical reflection, and inspire visions of more just future relations.

In Indigenous traditions, many stories are inextricably tied to the land, which suggests that location-based AR can be a powerful medium for sharing Native narratives, and perhaps social justice narratives more broadly [10–12]. "Why here?" is a guiding question for the Thámien Ohlone AR Tour: each of the three stops on the

walking tour is selected for its connection to its location [7]. The first stop showcases life-size 3D models of traditional tule reed houses at the historical Thámien Ohlone Village site, which lacks any physical commemoration. The second stop at Mission Santa Clara displays the names of Native individuals in the Chochenyo language around the church, highlighting its dual Catholic and Native significance [9]. The third stop, at the St. Ignatian lawn, depicts images from a tribal Pow Wow held annually, accompanied by a traditional Chochenyo healing song. Collectively, these three stops affirm the historical and ongoing presence of the Ohlone people.

The tour is co-designed by leaders of the Muwekma Ohlone Tribe and an interdisciplinary team of humanities and engineering faculty and students at Santa Clara University in the San Francisco Bay Area. Our co-design approach is guided by the principles of “design justice” [3], which extends human-centered design to intentionally center the voices of impacted communities and challenge oppressive structures and narratives. At the heart of this approach is the “nothing about us without us” principle pioneered by disability activists [2], which says that no design should proceed without the full and direct participation of the impacted communities. It is complemented by the notion of rhetorical sovereignty, which asserts the right of all Native peoples to articulate their own stories for their own purposes [6].

To realize this vision, we are innovating new co-design methods for AR experiences that link stories to specific locations [5]. A key method we have developed is “landmark-based affinity diagramming” (see Fig. 1 and Fig. 2) [11], which involves a three-step process that extends the “affinity diagramming” method that is already widely used in design [4, 8]. First, participants brainstorm ideas for stories and digital content for the AR experience, jotting these down on post-it notes. Second, these notes are clustered at different locations on a large physical map of the site, in this case, the Santa Clara University campus. Third, the different clusters, each representing an AR experience at a specific location, are discussed and prioritized. This method has proven effective in facilitating collaboration between diverse participants, including Native and non-Native people, those with technical expertise and those without, and individuals both familiar and unfamiliar with the campus landmarks. Crucially, it anchors the conversation in concrete locations, narratives, and media, contributing significantly to the actionable outcomes in the AR tour’s design and development.

In this project, the campus of Santa Clara University is leveraged as a ‘living laboratory’ for location-based AR for social justice. It offers sites of cultural significance for the Muwekma Ohlone Tribe, high foot traffic (about 100,000 annual visitors), and a safe walking space without vehicle traffic. This will allow us to develop additional co-design methods, improve the sustainability and scalability of the experience, and explore how it can inspire more just future relations. Building on our initial three stops, the Muwekma Ohlone Tribe and our team have envisioned about 15 additional stops around the campus.

The “We Are Still Here” video demonstrates the power of location-based AR for social justice. By tying stories to the land, AR experiences can show what has been here in the past, what is still here, and what could be here in the future.

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A IMAGES OF LANDMARK-BASED AFFINITY DIAGRAMMING



Figure 1: To create the Thámien Ohlone AR Tour at Santa Clara University, the team tried a new co-design exercise called “landmark-based affinity diagramming.” Here, members of the Muwekma Ohlone Native American Tribe discuss where to place a media asset on the map.



Figure 2: The team printed a 10 x 6 foot map of our historical campus, the site of the AR tour. Pink post-its are stories/messages the tribe wanted to share, e.g. “Hopes for a future Ohlone garden at the University.” Blue post-it notes are media assets, e.g. 3D models of tribal artifacts, recordings of songs, that were then displayed at that location in AR.